

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	19.4 / 4.1469
Time	2026-07-06 23:00 UTC
Location	29.8342, -95.4891
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 858	37.68 to 100; mean 72.5649	60.82 at 2026-07-06 22:59Z; dt 1 min
asos	temperature	degC	9 / 858	21.6667 to 36.6667; mean 29.0714	28.8889 at 2026-07-06 22:59Z; dt 1 min
asos	wind_direction	degree	9 / 1126	0 to 360; mean 138.33	240 at 2026-07-06 22:59Z; dt 1 min
asos	wind_speed	m/s	9 / 1164	0 to 15.4333; mean 2.1789	2.57222 at 2026-07-06 22:59Z; dt 1 min
noaa_gfs	gh_500	m	10 / 120	5891.13 to 5918.79; mean 5906.34	5898.52 at 2026-07-07 00:00Z; dt 60 min
noaa_gfs	pbl_height	m	10 / 120	19.6394 to 2153.73; mean 804.168	1236.03 at 2026-07-07 00:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	10 / 120	37.6598 to 51.5499; mean 45.0018	46.4419 at 2026-07-07 00:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	10 / 120	1004.8 to 1016.78; mean 1012.14	1010.33 at 2026-07-07 00:00Z; dt 60 min
noaa_gfs	t_850	degC	10 / 120	17.3771 to 21.3086; mean 19.196	19.4726 at 2026-07-07 00:00Z; dt 60 min
noaa_gfs	u_10m	m/s	10 / 120	-2.83889 to 3.62264; mean 0.4272	-0.0814392 at 2026-07-07 00:00Z; dt 60 min
noaa_gfs	v_10m	m/s	10 / 120	0.329297 to 4.9494; mean 2.1732	4.60966 at 2026-07-07 00:00Z; dt 60 min
openaq	ozone	ppm	17 / 1134	-0.004 to 0.069; mean 0.0178	0.026 at 2026-07-06 23:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 217	8 to 75; mean 22.2074	27 at 2026-07-06 23:00Z; dt 0 min
openaq	pm25	ug/m3	79 / 3131	0 to 83.6; mean 5.9986	17.2 at 2026-07-06 23:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 361	0 to 100; mean 37.3601	20 at 2026-07-06 23:01Z; dt 1.9 min
openweather	humidity	percent	5 / 361	50 to 95; mean 75.0305	65 at 2026-07-06 23:01Z; dt 1.9 min
openweather	precipitation	mm/h	5 / 361	0 to 5.62; mean 0.0445	0 at 2026-07-06 23:01Z; dt 1.9 min
openweather	pressure	hPa	5 / 361	1011 to 1018; mean 1015.2	1012 at 2026-07-06 23:01Z; dt 1.9 min
openweather	temperature	degC	5 / 361	23.78 to 36.13; mean 29.8207	33.02 at 2026-07-06 23:01Z; dt 1.9 min
openweather	wind_direction	degree	5 / 361	0 to 359; mean 194.028	186 at 2026-07-06 23:01Z; dt 1.9 min
openweather	wind_speed	m/s	5 / 361	0.45 to 6.09; mean 2.2045	5.03 at 2026-07-06 23:01Z; dt 1.9 min
purpleair	pm25	ug/m3	74 / 5107	0 to 372.3; mean 7.7545	6.5 at 2026-07-06 22:59Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_ch4_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0230724 to 0.0253077; mean 0.0244	0.0253077 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_hcho_column	mol/m^2	7 / 7	0.000201778 to 0.000235341; mean 0.0002	0.000218305 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_no2_column	mol/m^2	4 / 4	1.96909e-05 to 4.10314e-05; mean 0	4.08591e-05 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_o3_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_so2_column	mol/m^2	7 / 7	-4.80456e-05 to 4.72775e-05; mean 0	4.72775e-05 at 2026-07-06 19:12Z; dt 227.4 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-06 19:12Z; dt 227.4 min
tceq	co	ppm	3 / 187	0 to 1; mean 0.2267	0.4 at 2026-07-06 23:00Z; dt 0 min
tceq	no2	ppb	12 / 745	-2.3 to 35.8; mean 5.2601	19.4 at 2026-07-06 23:00Z; dt 0 min
tceq	so2	ppb	3 / 177	-0.5 to 19.3; mean 0.2593	-0.4 at 2026-07-06 23:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-05 06Z 92.18, 12Z 69.81, 18Z 47.96, 07-06 00Z 73.47, 06Z 88.28, 12Z 70.19, 18Z 55.49, 07-07 00Z 77.29, 06Z 89.88, 12Z 71, 18Z 57, 07-08 00Z 79.08, 06Z 91.83
asos	temperature	07-05 06Z 25.34, 12Z 30.5, 18Z 34.07, 07-06 00Z 28.88, 06Z 26.06, 12Z 30.32, 18Z 32.62, 07-07 00Z 28.31, 06Z 25.72, 12Z 29.83, 18Z 32.19, 07-08 00Z 25.82, 06Z 24.26
asos	wind_direction	07-05 06Z 66.67, 12Z 201.8, 18Z 203.9, 07-06 00Z 153.8, 06Z 119.4, 12Z 180.9, 18Z 166.7, 07-07 00Z 137.1, 06Z 73.4, 12Z 133, 18Z 148.1, 07-08 00Z 105.8, 06Z 28.18
asos	wind_speed	07-05 06Z 0.7395, 12Z 2.378, 18Z 2.71, 07-06 00Z 3.066, 06Z 1.243, 12Z 2.058, 18Z 3.818, 07-07 00Z 2.083, 06Z 0.7665, 12Z 1.424, 18Z 3.304, 07-08 00Z 2.915, 06Z 0.3747
noaa_gfs	gh_500	07-05 12Z 5907, 18Z 5917, 07-06 00Z 5905, 06Z 5907, 12Z 5892, 18Z 5902, 07-07 00Z 5898, 06Z 5905, 12Z 5899, 18Z 5914, 07-08 00Z 5912, 06Z 5918
noaa_gfs	pbl_height	07-05 12Z 163.7, 18Z 1779, 07-06 00Z 861.9, 06Z 263.3, 12Z 175.4, 18Z 1810, 07-07 00Z 1019, 06Z 227.9, 12Z 126.9, 18Z 1996, 07-08 00Z 976.4, 06Z 250.5
noaa_gfs	precipitable_water	07-05 12Z 42.4, 18Z 42.74, 07-06 00Z 43.05, 06Z 44.55, 12Z 41.89, 18Z 42.78, 07-07 00Z 46.68, 06Z 45.56, 12Z 46.22, 18Z 48.1, 07-08 00Z 48.87, 06Z 47.17
noaa_gfs	surface_pressure	07-05 12Z 1013, 18Z 1012, 07-06 00Z 1008, 06Z 1011, 12Z 1012, 18Z 1012, 07-07 00Z 1010, 06Z 1013, 12Z 1014, 18Z 1014, 07-08 00Z 1012, 06Z 1015
noaa_gfs	t_850	07-05 12Z 19.4, 18Z 18, 07-06 00Z 20.19, 06Z 20.15, 12Z 19.67, 18Z 17.97, 07-07 00Z 19.75, 06Z 19.78, 12Z 18.77, 18Z 17.85, 07-08 00Z 19.03, 06Z 19.77
noaa_gfs	u_10m	07-05 12Z 1.507, 18Z 1.102, 07-06 00Z 0.1826, 06Z 1.276, 12Z 1.021, 18Z 0.7598, 07-07 00Z -0.6914, 06Z 0.346, 12Z 0.8611, 18Z -0.1427, 07-08 00Z -1.34, 06Z 0.2456

Source	Metric	Values
noaa_gfs	v_10m	07-05 12Z 1.071, 18Z 1.399, 07-06 00Z 3.013, 06Z 2.565, 12Z 1.384, 18Z 1.541, 07-07 00Z 3.52, 06Z 2.189, 12Z 1.199, 18Z 1.606, 07-08 00Z 4.025, 06Z 2.566
openaq	ozone	07-05 06Z 0.00375, 12Z 0.01402, 18Z 0.03033, 07-06 00Z 0.02152, 06Z 0.007036, 12Z 0.01262, 18Z 0.03414, 07-07 00Z 0.01345, 06Z 0.005189, 12Z 0.01275, 18Z 0.03023, 07-08 00Z 0.02554, 06Z 0.004395
openaq	pm10	07-05 06Z 20.67, 12Z 18.72, 18Z 16.59, 07-06 00Z 15.89, 06Z 14.39, 12Z 30.22, 18Z 27.5, 07-07 00Z 15.89, 06Z 18.28, 12Z 33.17, 18Z 29.88, 07-08 00Z 21.94, 06Z 24.39
openaq	pm25	07-05 06Z 8.919, 12Z 6.443, 18Z 4.943, 07-06 00Z 5.454, 06Z 6.904, 12Z 6.452, 18Z 5.623, 07-07 00Z 4.88, 06Z 6.191, 12Z 6.682, 18Z 7.206, 07-08 00Z 4.586, 06Z 6.593
openweather	cloud_cover	07-05 06Z 84.67, 12Z 28.23, 18Z 92.41, 07-06 00Z 88.13, 06Z 41.86, 12Z 24.47, 18Z 24.9, 07-07 00Z 14.61, 06Z 4.033, 12Z 36.77, 18Z 46.57, 07-08 00Z 27.8, 06Z 4.68
openweather	humidity	07-05 06Z 90.83, 12Z 78.83, 18Z 57.69, 07-06 00Z 72.74, 06Z 86.38, 12Z 78.2, 18Z 59.53, 07-07 00Z 75.58, 06Z 87.8, 12Z 77.07, 18Z 59.23, 07-08 00Z 76.6, 06Z 89.88
openweather	precipitation	07-05 06Z 0, 12Z 0, 18Z 0, 07-06 00Z 0.006774, 06Z 0, 12Z 0, 18Z 0.005333, 07-07 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-08 00Z 0.5237, 06Z 0
openweather	pressure	07-05 06Z 1015, 12Z 1016, 18Z 1013, 07-06 00Z 1013, 06Z 1014, 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016, 12Z 1018, 18Z 1016, 07-08 00Z 1016, 06Z 1017
openweather	temperature	07-05 06Z 25.96, 12Z 29.88, 18Z 34.99, 07-06 00Z 30.2, 06Z 26.53, 12Z 29.63, 18Z 34.39, 07-07 00Z 29.5, 06Z 26.39, 12Z 29.93, 18Z 34.09, 07-08 00Z 27.7, 06Z 24.57
openweather	wind_direction	07-05 06Z 196, 12Z 216.4, 18Z 196.2, 07-06 00Z 175.9, 06Z 231.1, 12Z 233.5, 18Z 169.2, 07-07 00Z 178.6, 06Z 178, 12Z 214.6, 18Z 175.7, 07-08 00Z 170.7, 06Z 189.7
openweather	wind_speed	07-05 06Z 1.447, 12Z 1.85, 18Z 2.166, 07-06 00Z 2.95, 06Z 2.097, 12Z 1.838, 18Z 2.215, 07-07 00Z 2.186, 06Z 1.73, 12Z 1.439, 18Z 2.646, 07-08 00Z 3.145, 06Z 2.335
purpleair	pm25	07-05 06Z 13.53, 12Z 6.89, 18Z 5.56, 07-06 00Z 8.649, 06Z 9.184, 12Z 8.371, 18Z 6.384, 07-07 00Z 6.413, 06Z 7.879, 12Z 7.973, 18Z 8.328, 07-08 00Z 7.524, 06Z 9.442
sentinel5p	s5p_aer_ai_granule_available	07-05 18Z 1, 07-06 18Z 1, 07-07 18Z 1
sentinel5p	s5p_ch4_granule_available	07-05 18Z 1, 07-06 18Z 1, 07-07 18Z 1
sentinel5p	s5p_co_column	07-05 18Z 0.02308, 07-06 18Z 0.02528, 07-07 18Z 0.02495
sentinel5p	s5p_co_granule_available	07-05 18Z 1, 07-06 18Z 1, 07-07 18Z 1
sentinel5p	s5p_hcho_column	07-05 18Z 0.0002018, 07-06 18Z 0.0002254, 07-07 18Z 0.0002152
sentinel5p	s5p_hcho_granule_available	07-05 18Z 1, 07-06 18Z 1, 07-07 18Z 1
sentinel5p	s5p_no2_column	07-05 18Z 1.969e-05, 07-06 18Z 4.086e-05, 07-07 18Z 3.867e-05
sentinel5p	s5p_no2_granule_available	07-05 18Z 1, 07-06 18Z 1, 07-07 18Z 1
sentinel5p	s5p_o3_granule_available	07-05 18Z 1, 07-06 18Z 1, 07-07 18Z 1
sentinel5p	s5p_so2_column	07-05 18Z -4.805e-05, 07-06 18Z 4.559e-05, 07-07 18Z 2.628e-06
sentinel5p	s5p_so2_granule_available	07-05 18Z 1, 07-06 18Z 1, 07-07 18Z 1
tceq	co	07-05 06Z 0.2, 12Z 0.2, 18Z 0.2071, 07-06 00Z 0.21, 06Z 0.25, 12Z 0.2353, 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.25, 12Z 0.2167, 18Z 0.2706, 07-08 00Z 0.2, 06Z 0.2444
tceq	no2	07-05 06Z 4.217, 12Z 2.645, 18Z 2.04, 07-06 00Z 2.658, 06Z 4.488, 12Z 5.182, 18Z 4.274, 07-07 00Z 4, 06Z 6.039, 12Z 5.905, 18Z 7.493, 07-08 00Z 6.903, 06Z 10.96
tceq	so2	07-05 06Z -0.2333, 12Z 0.1706, 18Z 3.087, 07-06 00Z 0.11, 06Z -0.2, 12Z 0.1353, 18Z 0.075, 07-07 00Z -0.23, 06Z -0.2687, 12Z 0.3333, 18Z -0.05294, 07-08 00Z -0.2, 06Z -0.1867

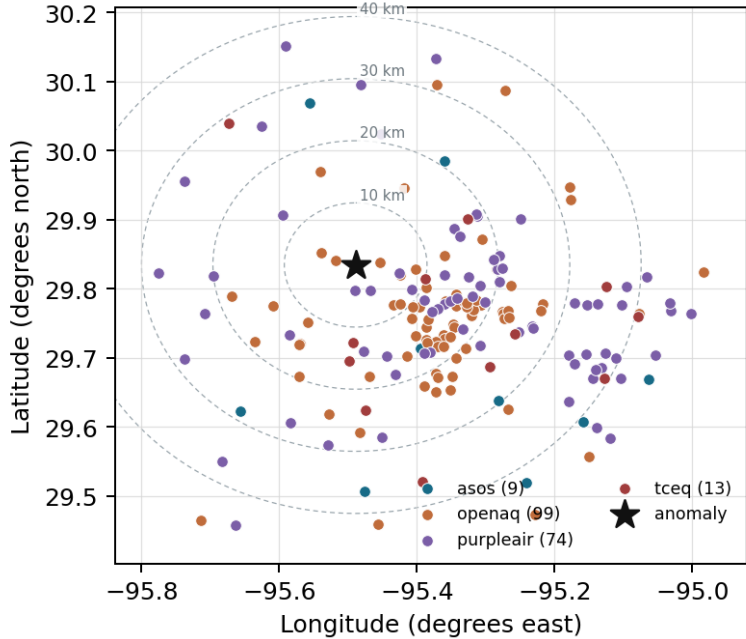
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

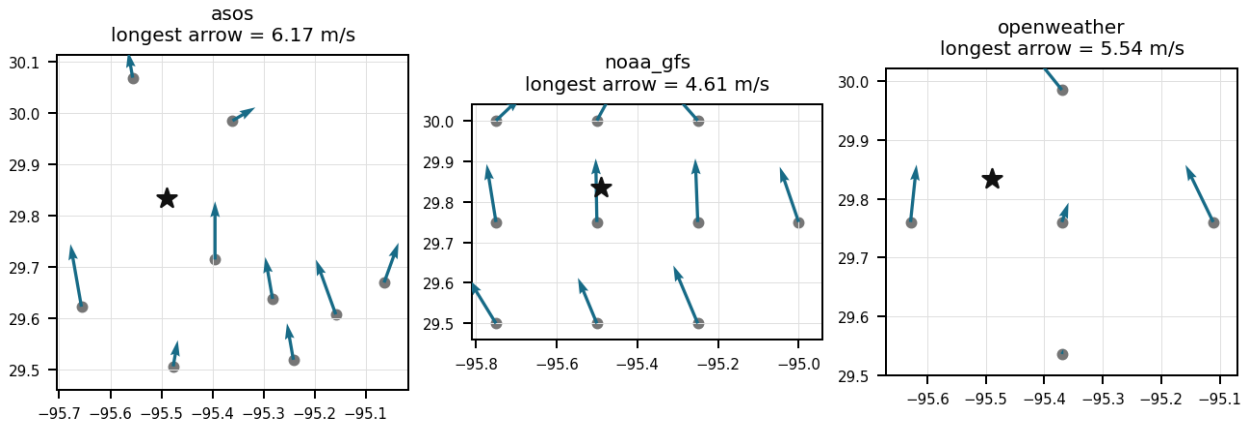
Source	Metric	Values
asos	humidity	MCJ (16.161 km) 70.94, IAH (20.788 km) 70.85, DWH (26.788 km) 76.55, SGR (28.573 km) 76.03, HOU (29.611 km) 71.15, AXH (36.503 km) 72.23, EFD (40.678 km) 70.4, LVJ (42.433 km) 71.77, +1 more
asos	temperature	MCJ (16.161 km) 29.35, IAH (20.788 km) 29.02, DWH (26.788 km) 27.86, SGR (28.573 km) 28.97, HOU (29.611 km) 29.76, AXH (36.503 km) 28.59, EFD (40.678 km) 30, LVJ (42.433 km) 29.24, +1 more
asos	wind_direction	MCJ (16.161 km) 130.6, IAH (20.788 km) 152.4, DWH (26.788 km) 109, SGR (28.573 km) 152.5, HOU (29.611 km) 172.4, AXH (36.503 km) 88.92, EFD (40.678 km) 151.4, LVJ (42.433 km) 119.6, +1 more
asos	wind_speed	MCJ (16.161 km) 2.714, IAH (20.788 km) 2.253, DWH (26.788 km) 1.456, SGR (28.573 km) 2.429, HOU (29.611 km) 2.848, AXH (36.503 km) 1.044, EFD (40.678 km) 3.016, LVJ (42.433 km) 1.756, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.50 (9.422 km) 5906, gfs:30.00,-95.50 (18.465 km) 5906, gfs:29.75,-95.25 (24.902 km) 5906, gfs:29.75,-95.75 (26.859 km) 5906, gfs:30.00,-95.25 (29.512 km) 5906, gfs:30.00,-95.75 (31.178 km) 5906, gfs:29.50,-95.50 (37.177 km) 5907, gfs:29.50,-95.25 (43.758 km) 5907, +2 more
noaa_gfs	pbl_height	gfs:29.75,-95.50 (9.422 km) 993.6, gfs:30.00,-95.50 (18.465 km) 1025, gfs:29.75,-95.25 (24.902 km) 939.3, gfs:29.75,-95.75 (26.859 km) 693.6, gfs:30.00,-95.25 (29.512 km) 1037, gfs:30.00,-95.75 (31.178 km) 592.2, gfs:29.50,-95.50 (37.177 km) 697.8, gfs:29.50,-95.25 (43.758 km) 638.4, +2 more
noaa_gfs	precipitable_water	gfs:29.75,-95.50 (9.422 km) 44.96, gfs:30.00,-95.50 (18.465 km) 45.55, gfs:29.75,-95.25 (24.902 km) 46.02, gfs:29.75,-95.75 (26.859 km) 43.59, gfs:30.00,-95.25 (29.512 km) 46.71, gfs:30.00,-95.75 (31.178 km) 43.76, gfs:29.50,-95.50 (37.177 km) 44.28, gfs:29.50,-95.25 (43.758 km) 45.17, +2 more
noaa_gfs	surface_pressure	gfs:29.75,-95.50 (9.422 km) 1012, gfs:30.00,-95.50 (18.465 km) 1011, gfs:29.75,-95.25 (24.902 km) 1014, gfs:29.75,-95.75 (26.859 km) 1011, gfs:30.00,-95.25 (29.512 km) 1012, gfs:30.00,-95.75 (31.178 km) 1009, gfs:29.50,-95.50 (37.177 km) 1013, gfs:29.50,-95.25 (43.758 km) 1014, +2 more
noaa_gfs	t_850	gfs:29.75,-95.50 (9.422 km) 19.2, gfs:30.00,-95.50 (18.465 km) 19.19, gfs:29.75,-95.25 (24.902 km) 19.07, gfs:29.75,-95.75 (26.859 km) 19.31, gfs:30.00,-95.25 (29.512 km) 19.06, gfs:30.00,-95.75 (31.178 km) 19.43, gfs:29.50,-95.50 (37.177 km) 19.19, gfs:29.50,-95.25 (43.758 km) 19.21, +2 more
noaa_gfs	u_10m	gfs:29.75,-95.50 (9.422 km) 0.6006, gfs:30.00,-95.50 (18.465 km) 1.105, gfs:29.75,-95.25 (24.902 km) 0.3956, gfs:29.75,-95.75 (26.859 km) 0.6981, gfs:30.00,-95.25 (29.512 km) -0.1486, gfs:30.00,-95.75 (31.178 km) 0.9456, gfs:29.50,-95.50 (37.177 km) 0.3089, gfs:29.50,-95.25 (43.758 km) 0.1539, +2 more
noaa_gfs	v_10m	gfs:29.75,-95.50 (9.422 km) 2.346, gfs:30.00,-95.50 (18.465 km) 1.783, gfs:29.75,-95.25 (24.902 km) 2.411, gfs:29.75,-95.75 (26.859 km) 2.353, gfs:30.00,-95.25 (29.512 km) 2.017, gfs:30.00,-95.75 (31.178 km) 2.146, gfs:29.50,-95.50 (37.177 km) 2.244, gfs:29.50,-95.25 (43.758 km) 2.148, +2 more
openaq	ozone	319 (0.008 km) 0.01656, 1319333 (11.294 km) 0.02247, 311 (15.421 km) 0.01674, 310 (17.39 km) 0.02217, 287 (18.775 km) 0.01846, 2046 (19.778 km) 0.01984, 312 (23.429 km) 0.01717, 325 (24.945 km) 0.005591, +9 more
openaq	pm10	15852419 (18.045 km) 24.32, 13380082 (24.981 km) 26.89, 271 (39.309 km) 15.39
openaq	pm25	10762809 (2.905 km) 9.999, 12337668 (3.368 km) 7.073, 14140750 (5.232 km) 6.798, 8967015 (6.377 km) 5.67, 8967363 (8.26 km) 7.465, 13787197 (8.381 km) 7.005, 8967413 (8.848 km) 6.558, 8967373 (10.454 km) 7.091, +71 more
openweather	cloud_cover	grid:center (14.139 km) 37.89, grid:west (15.756 km) 37.43, grid:north (20.333 km) 45.01, grid:south (35.126 km) 37.49, grid:east (37.388 km) 28.97
openweather	humidity	grid:center (14.139 km) 74.92, grid:west (15.756 km) 74.82, grid:north (20.333 km) 73.04, grid:south (35.126 km) 74.07, grid:east (37.388 km) 78.31
openweather	precipitation	grid:center (14.139 km) 0.08096, grid:west (15.756 km) 0.0175, grid:north (20.333 km) 0.005833, grid:south (35.126 km) 0.001944, grid:east (37.388 km) 0.116
openweather	pressure	grid:center (14.139 km) 1015, grid:west (15.756 km) 1015, grid:north (20.333 km) 1015, grid:south (35.126 km) 1015, grid:east (37.388 km) 1015
openweather	temperature	grid:center (14.139 km) 30.08, grid:west (15.756 km) 29.75, grid:north (20.333 km) 29.69, grid:south (35.126 km) 29.82, grid:east (37.388 km) 29.75
openweather	wind_direction	grid:center (14.139 km) 209.9, grid:west (15.756 km) 179, grid:north (20.333 km) 186.5, grid:south (35.126 km) 201.9, grid:east (37.388 km) 192.7

Source	Metric	Values
openweather	wind_speed	grid:center (14.139 km) 1.945, grid:west (15.756 km) 1.896, grid:north (20.333 km) 2.225, grid:south (35.126 km) 2.136, grid:east (37.388 km) 2.823
purpleair	pm25	51959 (4.098 km) 7.107, 301969 (4.533 km) 10.3, 155367 (6.247 km) 8.428, 223813 (8.787 km) 7.782, 302037 (11.11 km) 11.39, 246011 (12.486 km) 6.767, 130763 (12.96 km) 2.471, 296101 (13.109 km) 8.215, +66 more
tceq	co	482011052 (10.018 km) 0.3531, 482011035 (24.981 km) 0.1452, 482011039 (39.322 km) 0.177
tceq	no2	482010047 (0.0 km) 6.47, 482011052 (10.018 km) 11.32, 482011066 (12.524 km) 4.648, 482010055 (15.427 km) 2.612, 482010024 (17.385 km) 5.268, 482010416 (24.952 km) 4.292, 482011035 (24.981 km) 7.627, 482010029 (28.955 km) 1.992, +4 more
tceq	so2	482010051 (23.42 km) 0.7889, 482011035 (24.981 km) -0.02787, 482011039 (39.322 km) -0.03962

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 35

GFS model outputs indicate the planetary boundary layer height contracted from 1810 meters at 18:00 UTC to 1019 meters by 00:00 UTC on July 7, 2026, while ASOS stations reported light surface winds around 2.1 to 2.6 m/s that restricted vertical dispersion.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 2 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 3 of 35

Sentinel-5P satellite observations registered elevated tropospheric NO2 column densities over the Houston area on July 6, 2026 compared to July 5.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 4 of 35

TCEQ ground stations recorded a severe NO₂ concentration peak of 19.4 ppb at 2026-07-06T23:00:00+00:00, well above the local multi-station average of 5.26 ppb.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 5 of 35

The anomalous pollutant was nitrogen dioxide measured by a TCEQ monitor at 29.834, -95.489 in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 6 of 35

A regionally mixed photochemical or dust-driven event is contradicted because particulate matter remained modest near the anomaly, with OpenAQ PM_{2.5} at 17.2 ug/m³, PurpleAir PM_{2.5} at 6.5 ug/m³, and OpenAQ PM₁₀ at 27 ug/m³, while the NO₂ enhancement was highly localized at one TCEQ monitor relative to a window mean of 5.26 ppb rather than a broad multi-pollutant regional buildup.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 7 of 35

The accompanying chemistry is inconsistent with a regional photochemical or dust-dominated event because ozone near the monitor was only 0.026 ppm at the anomaly time and PM2.5 was not broadly elevated across nearby sensors.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 8 of 35

A shallow evening boundary layer combined with typical urban combustion emissions led to a sharp increase in ground-level NO2 concentrations measured by TCEQ monitors.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 9 of 35

A short-lived fresh combustion plume from nearby mobile or industrial sources is a likely cause of the 19.4 ppb NO2 spike because co-located surface NO2 was extremely elevated relative to expectation while ozone at the same time was only moderate, a pattern consistent with recently emitted NOx before photochemical aging converts NO to ozone and nitrate.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 10 of 35

Meteorological conditions favored persistence of near-surface NO₂ from local sources because the event occurred in warm, rain-free conditions with only moderate local wind speeds and a late-afternoon-to-evening transition toward a shallower boundary layer than daytime peak mixing, which would reduce dilution of fresh emissions.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 11 of 35

NOAA GFS model data and ASOS meteorological observations show that the evening shallowing of the planetary boundary layer combined with light winds led to surface trapping of NO₂ emissions.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 12 of 35

Sentinel-5P satellite observations detected elevated tropospheric NO₂ column densities of 4.086e-05 mol/m² over Houston on July 6, 2026, compared to 1.969e-05 mol/m² on the preceding day.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 13 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 14 of 35

Tropospheric NO2 column densities over Houston increased significantly on July 6 relative to July 5 as detected by Sentinel-5P, indicating a higher regional pollution burden.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 15 of 35

A fresh local combustion plume is supported because the TCEQ monitor recorded a severe NO2 spike of 19.4 ppb at 2026-07-06T23:00Z while colocated ozone was only 0.026 ppm and nearby TCEQ CO was 0.4 ppm, a combination consistent with recently emitted NOx that has not aged into an ozone-rich regional plume.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 16 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 17 of 35

The PM2.5 levels are elevated at 74 ug/m3, which is higher than the mean of 7.7545 ug/m3.

Mark exactly one:

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Note:

Claim 18 of 35

The anomaly is a result of a combination of factors including high temperatures, humidity, and stagnant atmospheric conditions, leading to an accumulation of pollutants in the area.

Mark exactly one:

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Note:

Claim 19 of 35

Sentinel-5P satellite observations over the region recorded a peak 6-hour mean NO2 column density of 4.086e-05 mol/m² on July 6, 2026, at 18:00 UTC.

Mark exactly one:

☐ V

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Note:

Claim 20 of 35

The air quality anomaly in Houston, Texas on July 6th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:

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Note:

Claim 21 of 35

Light southerly to southwesterly surface winds observed by ASOS and OpenWeather constrained urban dispersion and concentrated local nitrogen dioxide emissions near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 22 of 35

Transport from emission sources to the south or southwest of the monitor is a likely cause of the NO₂ anomaly because surface winds near the event were southerly to southwesterly, with ASOS reporting about 240 degrees and 2.57 m/s near 22:59 UTC and neighborhood weather analyses also showing southerly flow.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 23 of 35

Meteorology near the anomaly featured south-to-southwest low-level flow and only modest dispersion, which is consistent with transport of nearby urban or industrial combustion emissions toward the monitor.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 24 of 35

The NO₂ levels were also elevated in Houston, Texas on July 6th, with a peak value of 19.4 ppb at monitor 482010047.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 25 of 35

TCEQ ground monitors recorded a severe NO₂ concentration anomaly reaching 19.4 ppb at 23:00 UTC on July 6, 2026 in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 26 of 35

Transport from nearby urban or industrial sources is supported because surface winds near the event were southerly to southwesterly at about 2.6 to 5.0 m/s, with ASOS reporting 240 degrees and 2.57 m/s and OpenWeather reporting 186 degrees and 5.03 m/s near 23Z, which is consistent with advection of emissions toward the monitor from source areas to the south or southwest of the site.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 27 of 35

The PM2.5 levels exceeded the normal range in Houston, Texas on July 6th, with a peak value of 74 ug/m3 at sensor 51959.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 28 of 35

The TCEQ nitrogen dioxide anomaly at 2026-07-06T23:00:00+00:00 had a z-score of 6.1016079801547 and was classified as severe.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 29 of 35

The anomaly occurred on July 6th, around 23:01:52+00:00, and was located approximately 19.823 km from the query point.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 30 of 35

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 31 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 32 of 35

The TCEQ nitrogen dioxide measurement at 2026-07-06T23:00:00+00:00 was 19.4 ppb, compared with an expected value of 4.146896551724137 ppb.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 33 of 35

A severe thunderstorm or tornado event has stirred up pollutants and particles from the ground, contributing to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 34 of 35

At 2026-07-06T23:00:00+00:00, a TCEQ monitoring station at coordinates (29.834, -95.489) recorded an elevated NO2 concentration of 19.4 ppb against an expected baseline of 4.15 ppb, representing a severe z-score anomaly of 6.10.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 35 of 35

The Houston TCEQ monitor recorded a severe local NO2 spike of 19.4 ppb at 2026-07-06T23:00:00+00:00 versus an expected 4.15 ppb, which is consistent with a short-duration fresh combustion plume rather than typical local background conditions.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
